

Custom AV for Zoom Rooms: OSC API (Beta)

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Introduction

OSC control is available within the Custom AV Zoom Rooms Controller (CAVZRC) application on Windows or macOS. Please note that control commands are sent to the CAVZRC application, not to the Zoom Room device itself. This allows for control over multiple Zoom Room devices simultaneously and over the WAN.

This API is primarily intended for production use cases and integration with show control for automation. Unlike ZoomOSC, this API is focused on control actions and not meeting data outputs.

The OSC API is currently in Beta.

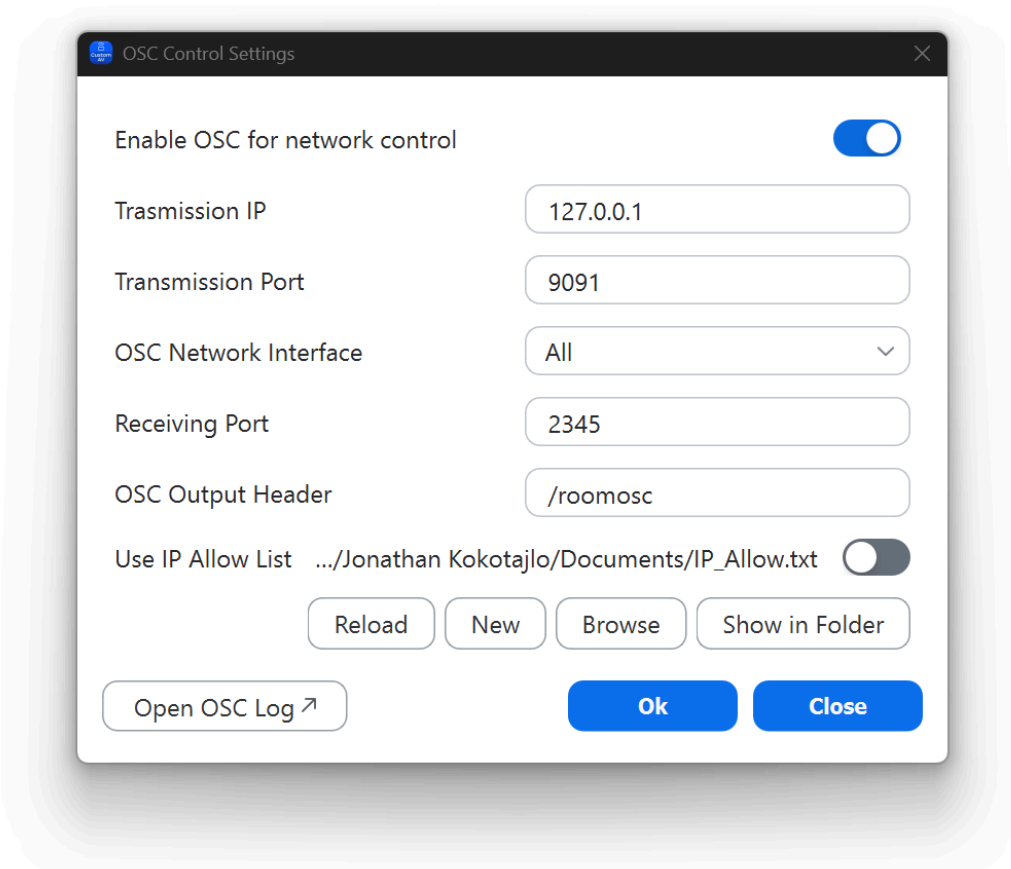
Requirements

Zoom Rooms version 6.5.0+

Custom AV Zoom Rooms Controller 1.2.0+

OSC Control Settings

Within the CAVZRC, you may adjust the network settings for OSC control by clicking on the gear icon in the top right of the application next to the "logout" button.



Enable OSC for network control	When enabled, the application will listen for incoming commands and output events
Transmission IP	The IP of the receiving device where the CAVZRC will send messages.
Transmission Port	The port of the receiving device where the CAVZRC will send messages.
OSC Network Interface	Select the network interface to use for incoming commands, or "All" for all interfaces.
Receiving Port	The port where the CAVZRC will listen for incoming messages.
OSC Output Header	This is the first part of the OSC address for output messages which can be customized to a string of your choice. The default is /roomosc.
Use IP Allow List	When enabled, you may load a .txt file containing a list of IP addresses, one per line. The CAVZRC will only allow control from these IPs on the selected interface(s).

Syntax Guidance

- This document outlines the Open Sound Control (OSC) syntax for use with the Custom AV for Zoom Rooms Controller (CAVZRC).
- OSC data is expected to be transmitted as UDP network traffic
- camelCase will be used in the documentation, but should not be enforced by the protocol, for example:
 - /zoomRooms and /zoomrooms will both be legal prefixes
 - ../roomId and ../roomid will both be legal target types
- OSC commands are implemented in multiple parts with the following documentation pattern:
 - An OSC address is the slash-delimited part of an OSC message
 - The OSC payload is the list of arguments that follow after the address
- A single whitespace will be used in the documentation to separate an OSC address from its payload argument(s)
- The use of "quotes" in the documentation may indicate a string parameter, but the quotes are not expected to be inside the actual strings received by the software
- We expect to allow UTF-8 characters in our strings
- The use of ../ indicates that there are unspecified components of the address before the shown component
- The use of ... indicates that there are unspecified arguments included in the payload
- If 'bool' is written in the documentation, it refers to an unsigned integer where 0 indicates False and 1 indicates True
- If 'str' is written in the documentation, it means a UTF-8 string
- In the command tables, the arguments in a payload will be separated by commas
- Sending a 'get' command instructs the CAVZRC that it should attempt to respond with the corresponding output for that command.

"Added" and "Paired"

- A Zoom Room (ZR) is considered "added" if it is selected by the CAVZRC to appear in the quick navigation tabs regardless of its pairing state
- A ZR is only considered "paired" if it is actively paired to the CAVZRC for control

Inputs

Inputs are commands received to the CAVZRC that are sent by 3rd party apps. These commands can either target specific Zoom Rooms or operate globally on the CAVZRC application itself.

All input command addresses must begin with /zoomRooms

Targeted Commands

Targeted Commands operate on specific ZR(s)

/zoomRooms/<targetType>/<room-command>

Target Types for Address

A target type specifies which format will be used to specify the Zoom Room(s) to operate the command upon. The valid target types are:

- ../roomID** - you will specify the ZR by its unique ID
- ../roomName** - you will specify the ZR by its display name
- ../roomIndex** - you will specify the ZR using its index in the list of ZRs added to CAVZRC
- ../allRooms** - the command will try to apply to ZRs added to the CAVZRC

Target Payload Arguments

Commands that use a target type in the address have certain requirements for the first argument in the payload of the OSC packet

- **Commands using ../roomID will use a string-type first argument**
Example: /zoomRooms/roomID/joinMeeting "DSKIJ3DG63125S" ...
- **Commands using ../roomName will use a string-type first argument**
Example: /zoomRooms/roomID/joinMeeting "Eyal's Room" ...
- **Commands using ../roomIndex will use an unsigned integer, one-indexed, as its first argument**
Example: /zoomRooms/roomID/joinMeeting 3 ...
- **Commands using ../allRooms will omit the target argument, so the command will begin with the command-specific arguments**
Example: /zoomRooms/roomID/joinMeeting ...

Global Commands

Global Commands operate on the CAVZRC itself and thus do not specify any Zoom Rooms.

/zoomRooms/<global-command>

Outputs

Outputs are commands sent by the CAVZRC to 3rd party receivers.

Format

By default, the CAVZRC will begin with /roomosc. However, the user can customize the header to any string that is legal for an OSC address, as they may be operating in an environment where multiple CAVZRCs are deployed and need address-level differentiation in their receivers.

Output Payload Prefix Arguments

Outputs that are specific to a Zoom Room are documented as having a prefix list of arguments containing multiple identifiers for the ZR that must appear before the output-specific arguments.

The output prefix is as-follows:

"roomId" "roomName" roomIdIndex ...

where roomId and roomName are strings and roomIdIndex is an unsigned integer representing the location in a one-indexed list of ZRs added to the CAVZRC

OSC Command List

Input Room Commands

Remember that all commands must take their target argument before the command-specific arguments are specified.

Address	Command Payload Arguments	Category	Description
../joinMeeting	str meetingID, str meetingPass, str userName	Join Flow	Join a meeting
../startMeeting		Join Flow	Start a meeting
../leaveMeeting		Join Flow	Leave a meeting
../setNDIContentOff	int channel_num	NDI	Set NDI Channel Content to Off
../setNDIContentParticipant	int channel_num	NDI	Set NDI Channel Content to Participant
../setNDIContentActiveSpeaker	int channel_num	NDI	Set NDI Channel Content to Active Speaker
../setNDIContentGallery	int channel_num	NDI	Set NDI Channel Content to Gallery
../setNDIContentScreenshare	int channel_num	NDI	Set NDI Channel Content to Screenshare
../setNDIContentSpotlight	int channel_num	NDI	Set NDI Channel to Spotlight Content
../setNDIContentPinGroup	int channel_num	NDI	Set NDI Channel to Pin Group
../setHWIOMode	int channel_num, int mode_index	HWIO	Set HWIO Mode
../setHWIOInputSelection	int channel_num, int video_index	HWIO	Set HWIO Channel to video at index
../setHWIOContentOff	int channel_num	HWIO	Set HWIO Channel Content to Off
../setHWIOContentTestSignal	int channel_num	HWIO	Set HWIO Channel Content to Test Signal
../setHWIOContentParticipant	int channel_num	HWIO	Set HWIO Channel Content to Participant
../setHWIOContentActiveSpeaker	int channel_num	HWIO	Set HWIO Channel Content to Active Speaker
../setHWIOContentGallery	int channel_num	HWIO	Set HWIO Channel Content to Gallery
../setHWIOContentScreenshare	int channel_num	HWIO	Set HWIO Channel Content to Screenshare

../setHWIOContentSpotlight	int channel_num	HWIO	Set HWIO Channel to Spotlight Content
../setHWIOContentPinGroup	int channel_num	HWIO	Set HWIO Channel to Pin Group
../setHWIOResolutionFrameRate	int channel_num, str resolution_framerate	HWIO	Set HWIO channel resolution and framerate
../getHWIOSupportedResolutionFrameRate	int channel_num	HWIO	Get HWIO channel resolution and framerate
../setHWIOAudioMix	int channel_num, int setting_index	HWIO	Set HWIO channel audio mix
../setDanteContentOff	int channel_num	Dante	Set Dante Channel Content to Off
../setDanteContentParticipant	int channel_num	Dante	Set Dante Channel Content to Participant
../setDanteContentMix	int channel_num	Dante	Set Dante Channel Content to Mixed Audio
../setDanteContentScreenshare	int channel_num	Dante	Set Dante Channel Content to Screenshare
../setNDIParticipantSelection	int channel_num, str exact_zoom_username	NDI	Select Participant for NDI Channel
../setNDIGallerySelection	int channel_num, int gallery_index	NDI	Select Gallery for NDI Channel
../setNDIScreenshareSelection	int channel_num, int screenshare_index	NDI	Select Screenshare for NDI Channel
../setNDIPinGroupSelection	int channel_num, int pin_group_index	NDI	Select Pin Group for NDI Channel
../setHWIOParticipantSelection	int channel_num, str zoom_username	HWIO	Select Participant for HWIO Channel
../setHWIOGallerySelection	int channel_num, int gallery_index	HWIO	Select Gallery for HWIO Channel
../setHWIOScreenshareSelection	int channel_num, int screenshare_index	HWIO	Select Screenshare for HWIO Channel
../setHWIOPinGroupSelection	int channel_num, int pin_group_index	HWIO	Select Pin Group for HWIO Channel
../setDanteParticipantSelection	int channel_num, str zoom_username	Dante	Select Participant for Dante Channel
../getNDIChannelConfig	int channel_num	NDI	Get the configuration of an NDI channel
../getHWIOChannelConfig	int channel_num	HWIO	Get the configuration of a HWIO channel
../getDanteChannelConfig	int channel_num	Dante	Get the configuration of a Dante channel
../getNDIChannelCount		NDI	Get the number of NDI channels

			available
../getHWIOChannelCount		HWIO	Get the number of HWIO channels available
../getDanteChannelCount		Dante	Get the number of Dante channels available
../setRoomMic	str mic_name	Device Settings	Set the microphone device for the room
../setRoomMainCamera	str camera_name	Device Settings	Set the primary camera for the room
../setRoomMultiCameraOn	str camera_name	Device Settings	Set a multi-camera device on for the room
../setRoomMultiCameraOff	str camera_name	Device Settings	Set a multi-camera device off for the room
../setRoomSpeaker	str speaker_name	Device Settings	Set the speaker device for the room
../getRoomMicList		Device Settings	Get a list of mic devices
../getRoomCameraList		Device Settings	Get a list of camera devices
../getRoomSpeakerList		Device Settings	Get a list of speaker devices
../muteMic		Device Settings	Mute the mic
../unMuteMic		Device Settings	Unmute the mic
../startCamera		Device Settings	Start the camera
../stopCamera		Device Settings	Stop the camera
../setNameTagAlignment	int location_index	Overlays	Set the name tag overlay location
../enableNameTagOverlay		Overlays	Enable the name tag overlay
../disableNameTagOverlay		Overlays	Disable the name tag overlay
../enableEmojiOverlay		Overlays	Enable the emoji overlay
../disableEmojiOverlay		Overlays	Disable the emoji overlay
../enableHandRaiseOverlay		Overlays	Enable the hand raise overlay
../disableHandRaiseOverlay		Overlays	Disable the hand raise overlay
../enableActiveSpeakerOverlay		Overlays	Enables the green active speaker border
../disableActiveSpeakerOverlay		Overlays	Disables the green active speaker border

../getOverlaySettings		Overlays	Get the overlay settings
../startDeviceShare		Content Share	Start sharing from a capture card
../startCameraShare	str camera_name	Content Share	Start sharing from a camera
../stopShare		Content Share	Stop sharing
../getRoomInfo		Room Info	Get information about the room
../getParticipantCount		Participant Info	Get the number of participants in the meeting
../activateCameraPreset	int preset_index	Camera Control	Activate the PTZ preset for the camera
../pairRoom		CAVZRC	Pair to the target Zoom Room
../unPairRoom		CAVZRC	Unpair from the target Zoom Room
../renameParticipant	string current_name, str new_name	Name	Change the name of the Zoom Room participant(s) in a meeting
../getSelectedPrimaryCamera		Device Settings	Get selected primary camera
../getSelectedMultiCameras		Device Settings	Get selected multi camera
../getSelectedMic		Device Settings	Get selected mic
../getSelectedSpeaker		Device Settings	Get selected speaker
../getMeetingStatus		Room Info	Get meeting status

Input Global Commands

Address	Description
/zoomRooms/getAddedRoomList	Get the list of rooms added to the CAVZRC
/zoomRooms/getPairedRoomList	Get the list of rooms paired to the CAVZRC
/zoomRooms/getAddedRoomCount	Get the number of rooms added to the CAVZRC
/zoomRooms/getPairedRoomCount	Get the number of rooms paired to the CAVZRC

Address	Command Payload Arguments	Category	Description
../supportedResolutionFrameRateHWIO	int channel_num, str supported_settings...	HWIO	Returns the supported resolutions/frame rates for a HWIO channels
../channelConfigNDI	int index, int status, str content, str selection	NDI	Returns the configuration of an NDI channel
../channelConfigHWIO	bool isActive, str channel_name, int mode (1 for output, 2 for input), int content (0 is not available, 1-X = index in dropdown), str selection (empty string if nothing, otherwise use the name), str resolution_fps, int audio_mix (0 means n/a, otherwise int index)	HWIO	Returns the configuration of a HWIO channel
../channelConfigDante	int index, int status, str content, str selection, str signal	Dante	Returns the configuration of a Dante channel
../channelCountNDI	int count	NDI	Returns the number of NDI channels available
../channelCountHWIO	int count	HWIO	Returns the number of HWIO channels available
../channelCountDante	int count	Dante	Returns the number of Dante channels available
../roomMicList	array of str	Device Settings	Returns a list of mic devices
../roomCameraList	array of str	Device Settings	Returns a list of camera devices
../roomSpeakerList	array of str	Device Settings	Returns a list of speaker devices
../roomOverlayConfig	int name_position, bool show_nametag, bool show_emoji, bool show_hand, bool show_border	Overlays	Returns the overlay settings For name position, 1 = left, 2 = center, 3 = right
../roomInfo	str availability, str room_type, str location, str web_zrc_url, str version	Room Info	Returns information about the room. Remember, the prefix information already shows the room's id, name, and index before the listed args appear

Output Room CommandsOutput Global Commands

Address	Command Payload Arguments	Description
../addedRoomsList	int max_list, int this_index, str roomID, str roomName	Returns the list of rooms added to the CAVZRC (one per room)
../pairedRoomsList	int max_list, int this_index, str roomID, str roomName	Returns the list of rooms paired to the CAVZRC (one per room)
../addedRoomsCount	int count	Returns the number of rooms added to the CAVZRC
../pairedRoomsCount	int count	Returns the number of rooms paired to the CAVZRC